

Real-time Detection of Acute Lymphoblastic Leukemia Cells Using Deep Learning

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Abstract— Acute lymphoblastic leukemia (ALL) is one of the most common types of cancer among children. It can rapidly become fatal within weeks, and as such, early diagnosis is critical. Problematically, ALL diagnosis mainly involves manual blood smear analysis relying on the expertise of medical professionals, which is error-prone and time-consuming. Thus, it is important to develop artificial intelligence tools that will identify leukemic cells from a microscopic image faster, more accurately, and cheaper. Here, we investigate the capabilities of a traditional convolutional neural network (CNN) and You Only Look Once (YOLO) models for real-time detection of leukemic cells. The YOLOv5s model shows 97.2% accuracy for the task of object detection of ALL cells, with the inference speed allowing 80 image frames to be processed per second. These new findings can provide valuable insight in applying real-time object detection algorithms for improving the efficiency of blood cancer diagnosis.

Keywords—Object Detection, Acute Lymphoblastic Leukemia, Real-time Detection, Convolutional Neural Network

I. INTRODUCTION

Acute lymphoblastic leukemia (ALL) is one of the most dangerous blood cancers known today. Primarily affecting children, ALL is one of the most common types of cancer among kids. Caused by the excessive proliferation of blood cells, ALL causes the bone marrow to produce immature lymphocytes, a type of white blood cell (WBC) that defends against infection and illness [1]. As a condition that can rapidly become deadly within weeks, early diagnosis is critical.

Problematically, leukemia diagnosis in its early stages has challenged researchers for years due to the cancer's initial mild symptoms and lack of tumors. Current manual approaches to blood smear analysis are time-consuming, expensive, and error-prone, relying on valuable resources like the expertise of trained medical professionals with specialized knowledge of image evaluation [2]. Beyond diagnosis, leukemia patients must endure hundreds of rounds of blood work for blood cell counts and analysis.

Since artificial intelligence has been increasingly applied to medicine, employing machine learning methods can encourage more efficient and precise diagnosis and blood work. Not only can increasing the speed of leukemia diagnosis and treatment reduce the suffering of thousands, but it can also help optimize the healthcare system, allowing doctors and specialists to treat more patients. Our objective is to identify the ALL cells in a blood smear quickly enough for real-time inference. For instance, given a large image of a group of cells, real-time inference could allow doctors to navigate the image while receiving an immediate response on each new section of the image. Exploring several neural networks, such as a traditional convolutional neural network (CNN) and You Only Look Once (YOLO) models, we strive to improve leukemia diagnosis and blood cell counting.

II. MATERIALS AND METHODS

A. Dataset

The data for this study comes from a public dataset called ALL-IDB, specifically designed for the evaluation of segmentation and image classification algorithms [3]. The microscopic images of blood samples in the dataset were captured with an optical laboratory microscope coupled with a digital camera. The dataset is split into two sections: first is the ALL-IDB1 dataset, which is designed for the purpose of training classification systems. The dataset also contains each image's blast cell centroid coordinates, if any, indicating the position of ALL lymphoblasts. We did not use the ALL-IDB2 dataset, a collection of cropped images from ALL-IDB1.

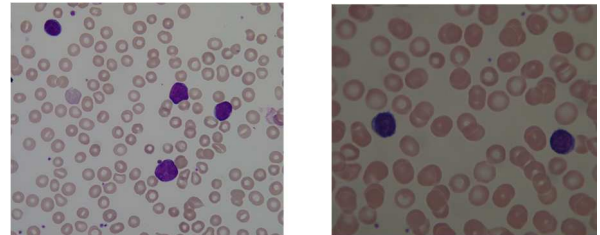


Figure 1: Positive (left) and negative (right) images from ALL-IDB1. Cancerous and healthy cells can be indistinguishable to the human eye.

In Figure 1, white blood cells are stained purple, while red blood cells are pink and more plentiful. For our ALL research, only the WBCs are relevant in determining whether the image is healthy or cancerous since those are the cells that can be immature. Before any models were trained, we resized all images due to different size ratios and computing resource constraints: we formatted the data to 900×1200 for the CNN and 1024×1024 for YOLO models.

B. Convolutional Neural Network

We began with a classification model aimed at classifying an entire image as healthy or cancerous. After cropping all data images to our target ratio, we separated 15% of the images from the IDB1 dataset for testing and left the remaining images for training. We then built a traditional CNN, which is designed to detect important features and learn adaptively. Since our goal was to reduce the runtime without sacrificing too much accuracy on high-resolution, we created a custom CNN instead of using a pre-trained model.

To compensate for a small dataset, we augmented the data by randomly flipping, rotating, and zooming in the images. In order to identify whether the image contained ALL cells, we refrained from augmenting the data by zooming. However, accuracy values did not change significantly if a few cells were cut out of the image, since all cancerous-labeled slides retained cancer cells to classify the whole image as positive.

Our CNN model consisted of 3 convolution-pooling layers followed by a dropout layer, a flatten layer, and 3 dense layers with one dropout layer in between. As we crafted the model, we inserted dropout layers between dense layers to prevent overfitting as we deepened the neural network. Model training was performed within 40 epochs, and we stopped adding layers once the training and validation losses stabilized together.

C. YOLO Models

The purpose of this study is to identify the leukemic cells in a blood smear image, so we explored object detection models to pinpoint the exact location of leukemic cells with bounding boxes. We employed transfer learning on a publicly available, pre-trained real-time object detection model. The selected models are the YOLO nets, which are known for their incredibly fast speed and advantages in real-time processing. The YOLO models were pre-trained on the COCO image dataset, and we implemented various YOLO versions for object detection on ALL cells [4].

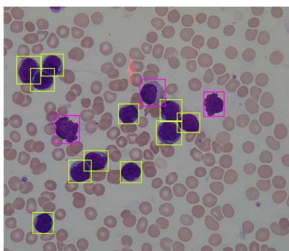


Figure 2: Annotated ground truth blood smear image. Leukemic and healthy WBCs are outlined by yellow and pink frames, respectively.

Using the coordinates from the IDB1 dataset, we annotated all cancer cells for the object detection model. Although the healthy cells were not our research focus, we also annotated all healthy WBCs.

For the transfer learning, we separated 15% of the images for the test dataset and inputted the remaining training images into the YOLOv5, YOLOv6, and YOLOv7 models, which are different versions of the YOLO model [5, 6, 7]. We explored the YOLOv5 more than the YOLOv6 and YOLOv7, given that the version 6 and version 7's training accuracies were far worse than the YOLOv5 on our ALL dataset—both more recent versions continuously identified a cell's cytoplasm as a completely separate cell.

Experimenting with version 5, we researched ALL identification using the YOLOv5 without initial weights and the YOLOv5s and YOLOv5m with pre-trained weights. These models differ in their sizes: the v5s, the small model, contained 213 layers and 7 million parameters, while the medium model had 290 layers and 20 million parameters. Finally, we tested the YOLOv5 model, which had no pre-trained weights but the same number of layers and parameters as the small model. Our objective in running this model was to determine whether pre-trained weights from normal, everyday objects would be beneficial in blood cell detection, a potentially very different dataset. Lastly, each YOLO model was trained within 150 epochs.

III. RESULTS AND DISCUSSION

On our test dataset, the CNN classification model had a 99.22% accuracy. However, it did not identify the location or number of cancer cells like the object detection model, making the CNN much less useful in practice. We focused more on the YOLO models, which could predict bounding boxes and classifications for each WBC cell.

Figure 3 indicates that the YOLOv5 is capable of reaching a high accuracy during the training process. The model without pre-trained weights performed notably worse than the YOLOv5s and YOLOv5m, likely because the pre-trained weights helped the model identify features and patterns on an abnormal image set as well.

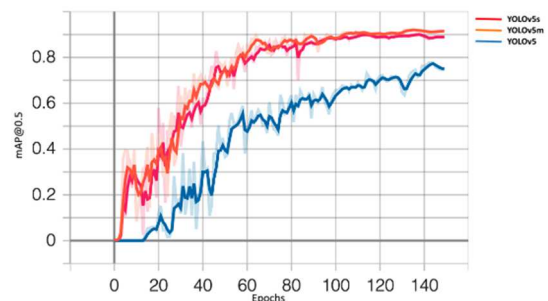


Figure 3: Training accuracy graph of all YOLOv5 models depicting that pre-trained weights performed better.

We then evaluated the various YOLO models using 17 images with 126 annotated WBCs to detect. The YOLOv5 models had the highest accuracy and speed.

Table 1: Performance of YOLO models in detecting ALL cells.

	YOLOv5s		YOLOv5m		YOLOv5		YOLOv6	YOLOv7	
Cancer(C) / Healthy(H)	C	H	C	H	C	H	C + H (All)	C	H
P	.885	.811	.936	.788	.679	.736	N/A	.554	.425
R	.985	.787	.955	.800	.970	.558	N/A	1.00	.814
mAP @0.5	.972	.831	.981	.853	.853	.703	.644	.737	.429
mAP @0.5: 0.95	.832	.627	.861	.654	.642	.401	.548	.608	.330
Inference Speed	12.3ms		21.9ms		22.0ms		16.0ms	42.7ms	

The metrics in Table 1 are indicators of the models' performances. The most significant metric is the mAP@0.5 for cancer cells, as we interpret it as the accuracy for the ALL cell location detection. The mAP@0.5:0.95 averages the mAP at each IoU threshold within the interval, but at higher thresholds like 0.95, the margin of error is unnecessarily stringent in practice for our uses; if the predicted bounding box is slightly shifted, a person reading the processed image will likely still be able to interpret which leukemic cell the model is indicating.

We note that the mAP for healthy cells is consistently lower than that of the cancer cells because the healthy cells were self-annotated without precise locations, leaving more room for errors in the ground truth boxes. All the same, our study focuses on identifying the cancer cells in an image, as finding ALL cells can be most useful for oncologists.

As depicted in Table 1, the YOLOv5s and YOLOv5m have significantly higher mAP@0.5 and faster runtimes than the other models. Such speed can be employed for real-time detection: for instance, the YOLOv5s only takes 12.3ms for each inference, meaning that a user could view the model's ALL identification on 80 unique frames of an image within one second. Finally, our computing machine was the Python 3 Google Compute Engine backend (GPU), for example a Nvidia Tesla P100, from Google Colab.

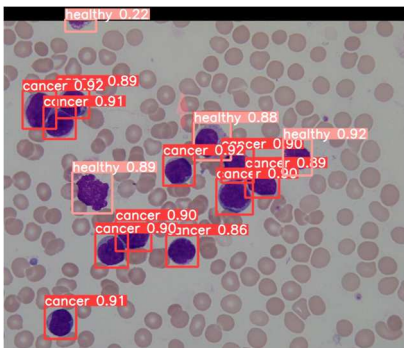


Figure 4: Resulting image from our YOLOv5s's object detection, including confidence levels.

IV. CONCLUSION

Based on the ALL-IDB dataset of microscopic blood smears, we propose to use the YOLOv5 model for real-time detection of leukemic cells. Specifically, we recommend the YOLOv5s. This smaller model performs inference with 97.2% mAP@0.5 accuracy and runs almost twice as fast as YOLOv5m without losing much accuracy. Our results suggest that object detection models can be more detailed, efficient, and practical than a traditional image classification. As well, the YOLO models could be implemented to report image level classification should such information be warranted. However, given our small dataset, we recognize that our findings should be validated with more field data from hospitals and research facilities. With professional healthy cell annotations, our models could also be modified to assist in blood cell counting in future efforts.

These new findings provide support to continue exploring the possibility of real-time object detection for ALL, which can improve ALL diagnosis and blood work. The YOLOv5 models' speed and accuracy provide great potential to improve healthcare productivity, and we believe that when implementation is achieved, our results will benefit a great number of people.

Code Availability: <https://github.com/echen053/all-cell-detection>

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